

Quoc Chuong NGUYEN

CURRICULUM VITAE

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GITHUB : <https://github.com/ChuongQuoc1413017>

RESEARCH INTEREST

Working on quantum algorithms and quantum machine learning, focusing on benchmarking, verification, and scalable simulation, with applications to quantum many-body systems and complexity analysis

Mathematics : Game Theory · Network Science · Category Theory
Computer Science : Complexity Theory · Property Testing · Machine Learning
Quantum Computation : Quantum Algorithm · Quantum Artificial Intelligence

PUBLICATION

Published

- **(Co-author) Chuong et al.** “FQsun: A Configurable Wave Function-Based Quantum Emulator for Power-Efficient Quantum Simulations”, IEEE Access, vol. 13, pp. 93271-93286, 2025 (updated Google Scholar’s citations: 6)
- **(First-author) Chuong et al.** “Qsun: an open-source platform towards practical quantum machine learning applications”, Machine Learning: Science and Technology, vol. 3, pp. 015034, 2022 (updated Google Scholar’s citations: 17)

Preprints

- (First-author) Chuong. “Network Sampling: An Overview and Comparative Analysis”, arXiv:2504.17701, 2025
- **(Co-author) Chuong et al.** “Hybridising Reinforcement Learning and Heuristics for Hierarchical Directed Arc Routing Problems”, arXiv:2501.00852, 2025
- (First-author) Chuong. “Monte Carlo Analysis of Boid Simulations with Obstacles: A Physics-Based Perspective”, arXiv:2412.10420, 2024
- **(First-author) Chuong and Trung Kien Le.** “Toward a comprehensive simulation framework for hypergraphs: a Python-base approach”, arXiv:2401.03917, 2024 (updated Google Scholar’s citations: 5)

ACADEMIC EXPERIENCE

Recent Remote Collaborations

- May 2024 - Now: In collaboration with Prof. Hy Truong Son (University of Alabama at Birmingham, US) and Prof. Thu Huong Dang (Lancaster University, England) on **Operations Research**

Reviewer for Peer-Reviewed Journals

- [SoftwareX](#), [Quantum Machine Intelligence](#), [Scientific Reports](#)

Research Positions

[Institute of Fundamental & Applied Sciences, Duy Tan University](#) July 2025 - Now

Website: <https://ifas.duytan.edu.vn/>

Position: Researcher

Job Description: develop quantum encoding frameworks for quantum many-body systems, focusing on benchmarking and simulation efficiency on GPU and quantum hardware

[QuantumLab - Ho Chi Minh City Institute of Physics, Vietnam](#)

Sep 2021 - May 2022

Website: <https://lantrann.github.io/QuantumLab-HCMIP>

Position: Affiliated Member

Job Description: develop quantum machine learning applications in chemistry and quantum metrology

Research Assistant for Dr. Hung Q. Nguyen, Vietnam

Nov 2019 - Apr 2023

Position: Research Assistant

Job Description: develop Qsun (Quantum Computing simulator) and Variational Quantum Eigensolver's applications in physics

EDUCATION

AUG 2023 - JUN 2025 **Master of Arts in Mathematics at SUNY, University at Buffalo**
(State University of New York at Buffalo), New York City, the US
Track: Applied Mathematics
CGPA: 3.889/4.0 and Degree GPA: 3.967/4.0 (A/A-: High Distinction)
Conferral Date: June 1, 2025

DEC 2014 - NOV 2018 **Bachelor of Science in Physics at University of Science (HCMUS)**
Vietnam National University, Ho Chi Minh City (VNU-HCM), VN
Track: Theoretical Physics
CGPA: 8.27/10 or 3.5/4.0 (A, top 2% over 300 students)
Graduated Seminar: Simulation of Quantum Many-body Dynamics using the Monte-Carlo algorithm with Python (Non-Thesis option)
Conferral Date: November 4, 2018

PROJECTS AND RESOURCES

Software

- QSun: Quantum platform to simulate quantum circuits using NumPy array
 - https://github.com/ChuongQuoc1413017/Quantum_Virtual_Machine
- Quantum Evolutionary Algorithm: Optimize quantum circuits' structure using Evolutionary Algorithms
 - https://github.com/ChuongQuoc1413017/Quantum_Evolutionary_Algorithm
- HyperRD: Simulation and analysis of random and dynamical hypergraph
 - https://github.com/ChuongQuoc1413017/Hypergraph_RD

Lecture Notes

- Summary of Lie Algebra in Particle Physics (Vietnamese)
 - <https://github.com/ChuongQuoc1413017/Note/blob/main/Lie%20Algebra.pdf>
- Quantum Machine Learning at $\langle$ QCS | 2021 $\rangle$
 - https://github.com/ChuongQuoc1413017/Note/blob/main/QML_Lecture_Note.pdf

TALKS/POSTERS AT CONFERENCES, WORKSHOPS, AND SCHOOLS

“Tutorial on Quantum Computing using Qiskit and Python” at School on Quantum Optimization Algorithms 2026, UPT, Vietnam	22 Apr - 25 Apr 2026 Lecturer
“Introduction to Quantum Algorithms” (Undergraduate Level) at Quantum Winter School, HCMUT, Vietnam	22 Nov 2025 Lecturer
“Quantum Machine Learning & Qsun platform” at PIPC 2025: Celebrating 100 years of quantum physics, Vietnam	13 Oct - 15 Oct 2025 Poster
“Quantum Machine Learning” at Quantum Computing Seminar, Vietnam	19 Aug 2025 Speaker
“Quantum Machine Learning” (Graduate Level) at Quantum Computing School - $\langle$ QCS 2021 $\rangle$, Vietnam	04 Dec - 12 Dec 2021 Contributory Lecturer
“Half-wormhole solutions & Black Hole singularity” at The 46th Vietnam Conference on Theoretical Physics (VCTP-46)	13 Oct 2021 Poster

HONORS & AWARDS

Fellowships & Teaching Assistantship (SUNY, University at Buffalo) Description: for Master/Ph.D. student with a good profile (GPA at least 3.0/4.0, at most 10 students selected) each semester	Aug 2023 - June 2025 2 academic years
Academic incentive fellowships (University of Science, HCMUS) Description: for students with the highest GPA (top 10 students) and contributions to society through activities each semester	Nov 2014 - Nov 2018 4 academic years

WORKING EXPERIENCE

Teaching

Teaching Assistant (TA) at HCMUS, Vietnam (Graduate Level) Description: Instructor in Physical-Informed Neural Networks class	Sep 2025 - Dec 2025
Mentor of Directed Reading Project program at SUNY, UB Description: teach undergrad students about Computational Thinking and Data Structures & Algorithms	Jan 2024 - May 2024
Teaching Assistant (TA) at SUNY, UB (Undergraduate Level) Description: TA in Calculus, Linear Algebra, and Stochastic Processes	Aug 2023 - May 2025

Company

FUNiX Co. Ltd., Vietnam

Jul 2020 - Nov 2021

Position: Collaborative Assistant

Job Description: develop machine learning & data science courses, translate course content from Vietnamese to English, and fix errors

Saigon Scientists Co., Vietnam

Jul 2018 - Jul 2019

Position: Research & Development Assistant

Job Description: develop STEM lessons, translate courses' contents from Vietnamese to English, and project management

LANGUAGES & COMPUTER SKILLS

VIETNAMESE:	Mother tongue
ENGLISH:	Fluent (professional working proficiency)
PROGRAMMING LANGUAGES:	Proficient: Python, NetworkX, Qiskit, PennyLane Basic: Mathematica, Matlab, R, LaTeX, Haskell, Lean

SELECTED ACADEMIC EVENTS ATTENDED

CQT Quantum Hackamonth	1 - 26 Jun 2026
VIASM-ICTP Summer School in Group Theory and Representation Theory	9 - 21 Aug 2022
Topological Quantum Electrons Interacting In-persons	10 - 16 July 2022
Vietnam - USA Joint Mathematical Meeting - VNUS 2019	10 - 13 Jun 2019

SELECTED CERTIFICATIONS

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- [Introduction to Discrete Mathematics for Computer Science Specialization](#) on *Mathematical Thinking, Probability, Combinatorics, and Graph Theory* at [Coursera](#) in 2022, Credential ID: ZDLRFFSJ6H8N
 - [Introduction to Logic and Critical Thinking Specialization](#) on *Argument Analysis, Deductive Reasoning, Inductive Reasoning, and Fallacies* at [Coursera](#) in 2022, Credential ID: PNT6RQ3G9Q89
 - [Mind and Machine Specialization](#) on *Cognitive Science and Artificial Intelligence* at [Coursera](#) in 2022, Credential ID: HNTZFND54LU7

REFERENCES

Prof. Thu Huong Dang (Google Scholar) Department of Management Science Lancaster University, England Email: t.h.dang@lancaster.ac.uk	Prof. Long Tran-Thanh (Google Scholar) Department of Computer Science University of Warwick, England Email: long.tran-thanh@warwick.ac.uk
Dr. Hung Q. Nguyen (Google Scholar) Nano and Energy Center University of Science, VNU Hanoi, Vietnam Email: hungngq@hus.edu.vn	Dr. Lan Tran (Google Scholar) Department of Physics University of Science, HCMC, Vietnam Email: tnlan@hcmus.edu.vn
Prof. Le Ho (Google Scholar) QuantumLab-HCMIP Tohoku University, Japan Email: binho@fris.tohoku.ac.jp	Prof. Hy Truong Son (Google Scholar) Department of Computer Science University of Alabama at Birmingham, US Email: thy@uab.edu